Systems of Equations — Number of Solutions

Practice Paper — Digital SAT Style

Difficulty Mix: Easy (Q1–4) | Medium (Q5–9) | Hard (Q10–12) · Total: 12 Questions

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DIFFICULTY: EASY

Q1. How many solutions does the following system have?

$y = 3x + 5$

$y = 3x - 2$

A) 0
B) 1
C) 2
D) Infinitely many

DIFFICULTY: EASY

Q2. Which system of equations has infinitely many solutions?

A) $y = 2x + 1$ and $y = 2x + 3$
B) $y = 2x + 1$ and $y = -2x + 1$
C) $y = 2x + 1$ and $4y = 8x + 4$
D) $y = 2x + 1$ and $y = 3x + 1$

DIFFICULTY: EASY

Q3. How many solutions does the following system have?

$y = 2x + 1$

$y = -x + 7$

A) 0
B) 1
C) 2
D) Infinitely many

DIFFICULTY: EASY

Q4. For the system $ax + 2y = 6$ and $3x + y = 4$, what value of a makes the system have no solution?

A) $a = 3$

B) $a = 6$

C) $a = 1.5$

D) $a = -6$

DIFFICULTY: MEDIUM

Q5. The system of equations below has no solution.

$4x - 6y = 10$

$2x - 3y = k$

What value of k makes this true?

A) $k = 5$

B) $k = 4$

C) $k = 6$

D) Any value of k except 5

DIFFICULTY: MEDIUM

Q6. How many solutions does this system have?

$x^2 + y = 5$

$y = -x + 3$

A) 0

B) 1

C) 2

D) Infinitely many

DIFFICULTY: MEDIUM

Q7. The system $2x + ky = 8$ and $6x + 12y = 24$ has infinitely many solutions.

What is the value of k?

A) 2

B) 3

C) 4

D) 6

DIFFICULTY: MEDIUM

Q8. For the system $px + 4y = 12$ and $x + 2y = 6$, which value of p gives the system infinitely many solutions?

- **A)** $p = 1$
- **B)** $p = 2$
- **C)** $p = 3$
- **D)** $p = 4$

DIFFICULTY: MEDIUM

Q9. The system $y = x^2 - 4$ and $y = 2x - 1$ intersects at how many points?

- **A)** 0
- **B)** 1
- **C)** 2
- **D)** 3

DIFFICULTY: HARD

Q10. In the system $cx + 2y = 4$ and $2x + cy = 6$, for what value(s) of c does the system have NO solution?

- **A)** $c = 2$ only
- **B)** $c = -2$ only
- **C)** $c = 2$ or $c = -2$
- **D)** $c = 4$

DIFFICULTY: HARD

Q11. The system $kx^2 - 3y = 9$ and $y = kx - 3$ has exactly one solution when the line is tangent to the parabola. For $k = 3$, how many solutions does the system have?

- **A)** 0
- **B)** 1
- **C)** 2
- **D)** Infinitely many

DIFFICULTY: HARD

Q12. The system below has solutions when c is in a certain range.

$x + y = c$

$x^2 + y^2 = 9$

For how many integer values of c does the system have at least one solution?

A) 3

B) 5

C) 7

D) 9